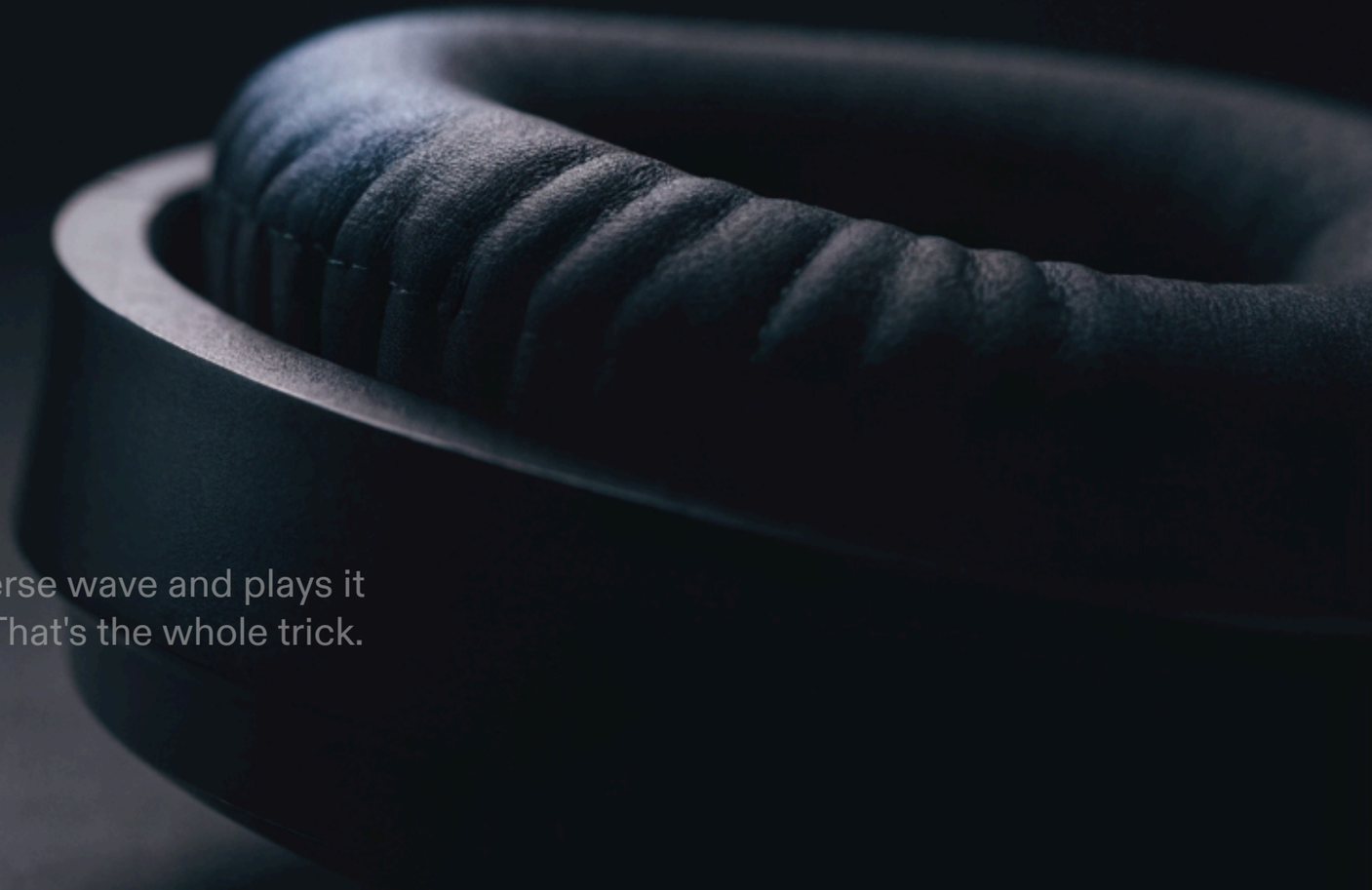
Silence is Engineered

What the DSP in your ANC headphones actually does — and why it's hard.

Noise just... disappears.

A mic captures the noise. A chip computes the exact inverse wave and plays it back — in under 10 milliseconds. The two waves cancel. That's the whole trick.



Some numbers worth knowing

48,000 samples/sec
How often the ADC samples the mic signal

<10 ms
Total pipeline latency budget, mic to speaker

45 dB
Best-in-class noise reduction (AirPods Pro 2)

20–1000 Hz
Frequency range where ANC is effective

2–6 mics
Per earbud in a hybrid ANC system

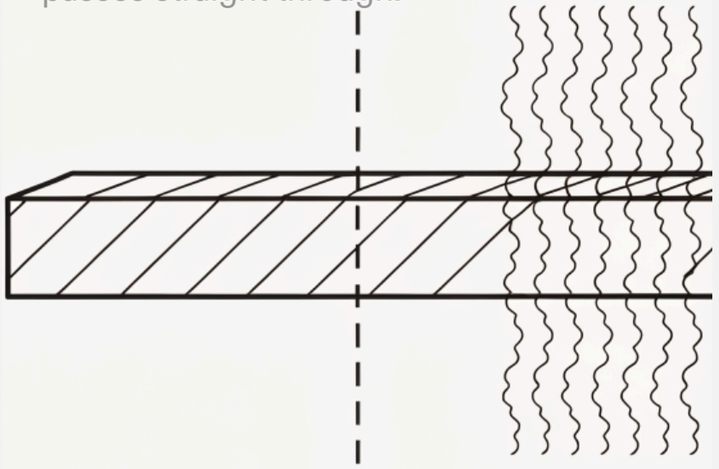
300 MHz+
DSP clock speed needed to run the pipeline

Specs based on publicly available data for flagship consumer ANC earbuds, 2023-2024

Passive and Active

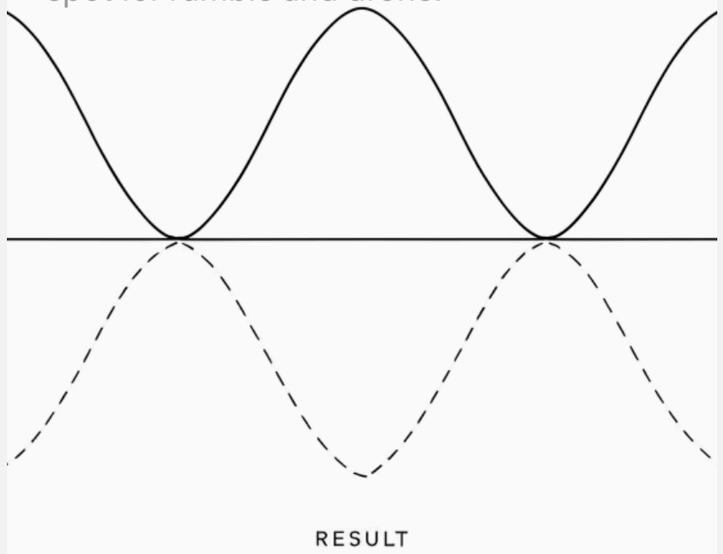
Passive

Physical seal only. Foam and ear cups block sound above ~1000 Hz. No electronics. Low-frequency rumble passes straight through.



Active (ANC)

Mic captures incoming sound. DSP inverts it. Speaker plays the anti-noise signal. Cancels 20–1000 Hz — the sweet spot for rumble and drone.



How it works

Every time you press ANC...

...this is what runs in under 10 milliseconds

1 Mic hears noise

A feedforward mic on the outside captures incoming noise before it reaches your ear.

2 ADC digitises it

The analog sound wave is converted to numbers — 48,000 samples every second.

3 DSP inverts it

The chip computes the exact opposite waveform using an adaptive filter (LMS algorithm). This is the hard part.

4 DAC converts back

The digital anti-noise is turned back into an analog electrical signal.

5 Speaker plays it

Anti-noise plays at your ear alongside your music. The two waves cancel via destructive interference.

All five steps complete in under 10ms — the threshold where your brain would notice the delay.

Quick — what even is sound?

Sound is moving air. Molecules get pushed and pulled in a wave — your eardrum vibrates, and your brain hears it as sound.

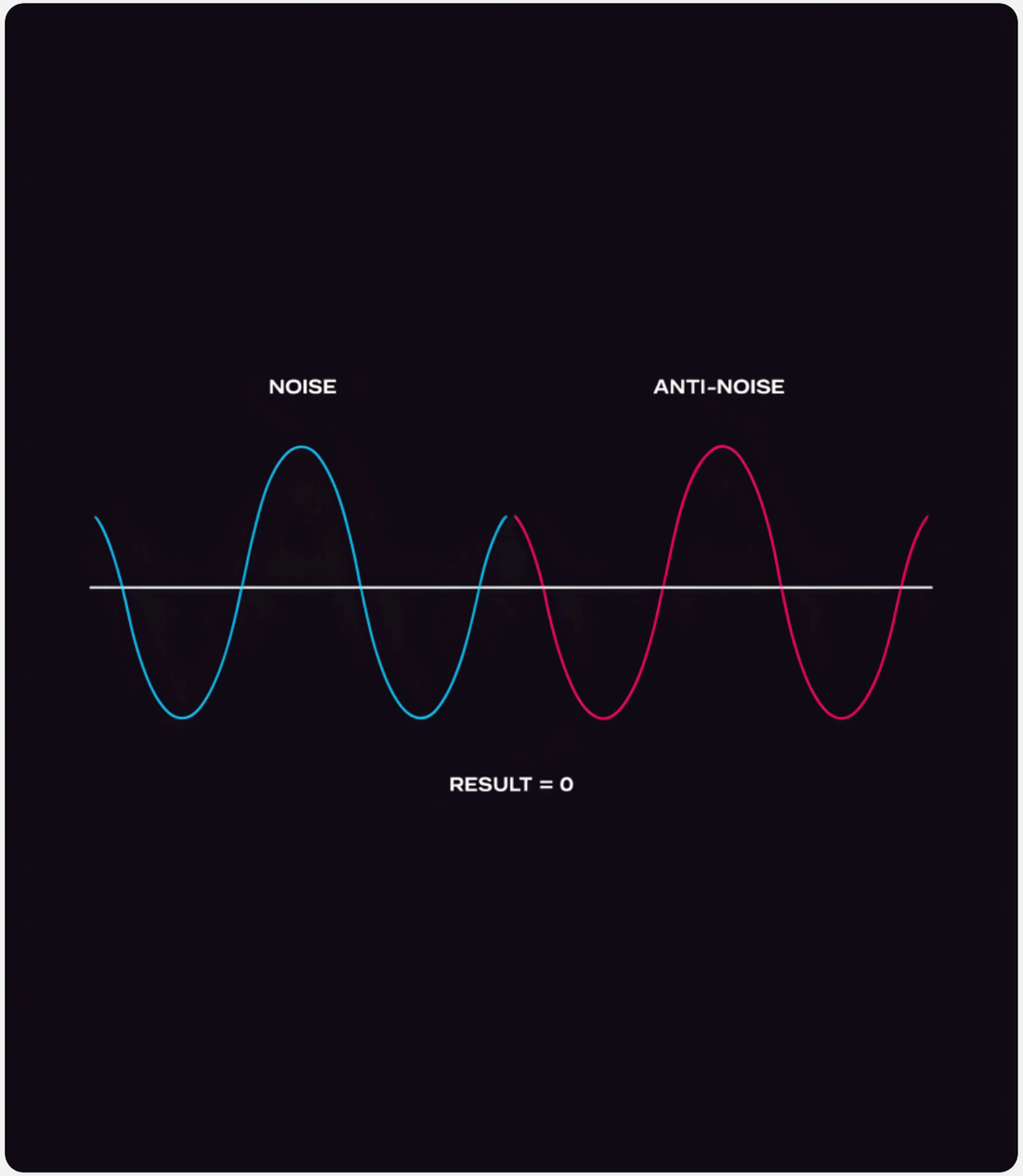
Three things describe any wave:

Frequency — how high-pitched it sounds (Hz)

Amplitude — how loud it is

Phase — where in the cycle it currently is

ANC only cares about one of these — phase.

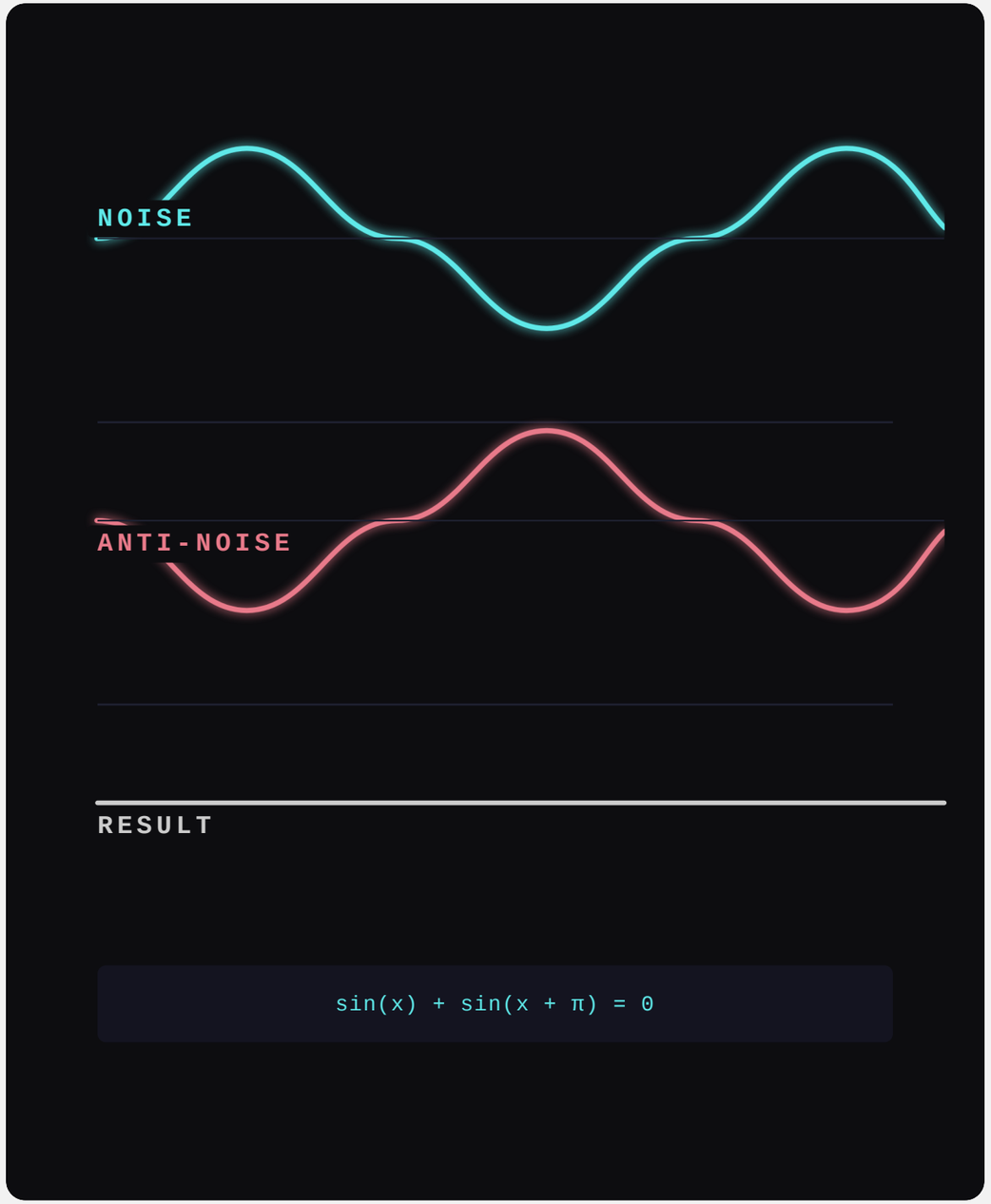


Why do opposite waves cancel?

NOISE	Pressure wave from the world around you
ANTI-NOISE	Inverted copy from the DSP chip
RESULT	They add to zero. Your ear hears nothing.

$$\sin(x) + \sin(x + \pi) = 0 \text{ for all } x$$

That's the whole thing, right there.



There are two ways to do this — and both have problems

Feedforward

Mic: outside ear cup

Gets a head start on noise
Hears noise before it reaches you. Reacts quickly, works well at higher frequencies.
The catch: no self-correction

1. No loop to verify the output
2. Filter error = uncorrected cancellation
3. Works best when noise is predictable

Feedback

Mic: inside cup, near ear

Corrects what actually reached your ear
Self-correcting loop, better at low frequencies. Closes the gap between intent and reality.
The catch: oscillation risk

1. Output re-enters the input signal
2. Wrong gain causes oscillation
3. High-pitched whine? This is the culprit

Why the filter adapts in real time

Fixed filters don't work. The world changes constantly, so the filter rewrites its own weights to match — 48,000 times per second using Least Mean Squares.

Capture Error

Measure residual error — the noise that leaked through the filter

Step 1

Nudge Weights

Adjust filter weights using error and reference input to reduce noise

Step 2

Repeat Continuously

Run the update 48,000 times per second to adapt in real time

Step 3

It's harder than it looks — here's why

The latency wall

Mic → filter → speaker must finish in under 10ms. Any slower and you hear a phase-shifted echo.

Wind

Turbulent pressure on the mic isn't a real sound wave. The filter can't cancel it — on a run, this ruins everything.

High frequencies

Short wavelengths move too fast. The anti-noise arrives after the wave has already passed your eardrum.

Music distortion

Anti-noise processing can subtly colour the audio. Making it sound transparent is as much art as engineering.

Temperature

Speed of sound shifts with temperature. A filter calibrated at 22°C is measurably off at 5°C.

Battery

The DSP never sleeps. Full ANC costs 4-6 hours of battery life. Chip efficiency is a constant arms race.

Who's doing this best right now

	Apple — H2 chip	Sony — QN1e / HD QN1	Bose — BSAC chip
Architecture	1. Custom DSP, 2 mics per bud 2. Up to 45 dB noise reduction 3. Single chip handles ANC and audio	1. Dedicated ANC processor, separate from audio path 2. 32-bit signal processing 3. No cross-interference between ANC and audio	1. 6 microphones per earbud 2. High mic density for broad-spectrum capture
Standout feature	1. Transparency mode uses the same mics in reverse — same hardware, flipped logic	1. Handles ANC and audio enhancement simultaneously — no compromise between the two	1. QuietMode maps your ear canal geometry, tuning ANC coefficients to your specific ear shape

That's not magic. It's signal processing.

And next time it doesn't work - check if you're running in the wind.